



The startup's corporate and R&D hub in Sonipat is spread over approx. 1850 square metres (20,000 square feet). This headquarter is the focal point for designing, developing, testing, and evaluating products. Both corporate and sales/marketing teams are stationed here. Its production facility, roughly 25 kilometres away, spans a 1-acre manufacturing plant.

"We have invested around 300 million from inception to date, covering design, development, infrastructure, and plant facilities over nearly four years. Our design, development, and testing equipment include 3D printers, prototype testing machines, EB simulators, LCR meters, oscilloscopes, and current probes. Additionally, we utilise software like SolidWorks, Catia, and Altair for PCB and firmware design," shares Mayank.

While discussing challenges,



Mayank Jain, Founder & CEO, E-Fill Electric

Mayank shares that acquiring the right resources is challenging, as the market is relatively new, and expertise is scarce. Another hurdle is finding suitable vendor partners due to competition, especially from Chinese OEMs, and vendors' reluctance to support low volumes. Addition-

ally, financing poses a challenge for chargers and three-wheelers, with limited options available, leading to higher interest rates.

The startup is developing a 360-kilowatt DC fast charger for the US market and seeking vendor partners capable of providing UL-certified products or components. "Our search spans vendors from Germany, China, and Korea as we explore multiple partnerships," adds Mayank.

The startup has secured nearly six acres of land for a new manufacturing facility. Operations are projected to commence there in approximately one and a half years.

The company's revenue surged from 135 million to 280 million in the previous fiscal year. In the next two years, the founder expects a fivefold increase!

—Nitisha Dubey

2 CLEANING AND COOLING THE AIR WITH KARBAN AIRZONE

An air-purifier, a fan, and lights—all in one modern chandelier-style appliance! Karban's bladeless fans, Karban Airzone, with air purification systems offer just that.

Jaipur-based startup Karban Envirotech Pvt Ltd is transforming the consumer appliance market with its bladeless fans that have built-in air purification and lighting systems.

Founded in December 2021, Karban, named after founder Karan Bansal, has designed and manufactured a bladeless ceiling fan with a built-in air purifier and lights. The startup has a unique design that helps reduce electricity consumption and provides multi-utility air flow, air purification, and lighting for homes, offices, hotels, and restaurants, among others.

In October 2021, Bansal left his cushy job at a company that designs computational fluid dynamics (CFD)

software in Wisconsin to evaluate the growing air pollution problem back home.

Initially focused on outdoor air pollution solutions, Bansal soon realised the limitations of this approach and shifted his focus to indoor air pollution solutions, coming up with the design of a bladeless ceiling fan. Within 18 months, Bansal and his co-founder Tanya Goyal were ready with the final product and began commercialisation with initial sales to friends, family, and website visitors for testing and feedback collection.

The product, Karban Airzone, integrates multiple functions (air purification, fan, lighting) into one product, aiming to reduce material, energy, and packaging costs of multiple devices.

It addresses indoor air pollution continuously while providing functionality as a fan. The product targets both households and commercial settings, offering safety benefits and space optimisation.

Bladeless fans, though uncommon, are not new. With a mixed-flow impeller or a crossflow blower to draw air from one side and discharge it from the other, the blower's blades rotate, creating a vortex within itself. This pulls air from one side and expels it from the other, creating a directional airflow system, enclosed within a casing.

"Since we already had airflow coming in and out, creating a low-pressure environment within the product, we strategically added a filter